



Guglielmo Gemignani

Head of Software

PROFILE

A robotics and AI expert with 15+ years of experience and more than eight years leading engineering teams. Currently leads the software department of a startup building uncrewed aerial systems, covering drone software and Ground Control Stations, from architecture and autonomy to simulation, software QA, and compliance with aerospace standards. Holds a PhD in Robotics and Artificial Intelligence, with 20+ academic publications and four patents.

WORK EXPERIENCE

Head of Software

Jul 2026 – Present

Twentyfour Industries | Munich

- Lead the software department of a startup building uncrewed aerial systems.
- Responsible for drone software and Ground Control Stations, from architecture and autonomy to simulation and software quality assurance.
- Own compliance with aerospace standards, and hire and grow the engineering organization.

Keywords: UAS, Ground Control Stations, Autonomy, Simulation, SW QA, Aerospace Standards, Hiring

Head of Robotic Integration

Mar 2022 – Jul 2026

Magazino – a Jungheinrich Company | Munich

- Disciplinary supervisor and technical lead for a team of 20+ developers across two companies.
- Responsible for robot architecture design, behavior development, robot simulators, and software and hardware quality assurance.
- Focus on developing high-level software for autonomous robots, including [TORU](#), [SOTO](#), and [Jungheinrich's EAEa](#).
- Managed and enhanced a commercial Behavior Tree-based framework, adopted by multinational companies for autonomous robot behavior development, contributing to the successful acquisition of the company.

Keywords: SCRUM, SAFe, Product Owner, Product Requirements, Behavior Trees, VDA5050, SW & HW QA, ROS, Unity, Gazebo, GitLab, Docker, Testing Strategy

Team Lead Behaviors & Reasoning

Jan 2018 – Mar 2022

Magazino GmbH | Munich

- Supervised a multidisciplinary team, specializing in robot behavior development, data analysis, and software QA.
- Directed the team's contributions to the successful development and launch of three innovative products: [TORU](#), [SOTO](#), [ACROS.AI](#).
- Ensured the delivery of high-quality software through robust testing, analysis, and agile development practices.

Keywords: Behavior trees, Automated planning, SW QA, Gazebo, Buildbot, Data Analysis, Jupyter, GCP.

CONTACT



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[g-gemignani.github.io](https://github.com/g-gemignani)



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EDUCATION

PhD Visitor

Carnegie Mellon University

CORAL Laboratory, 2014–2015

PhD in Robotics and Artificial Intelligence

Sapienza, University of Rome

Thesis: Acquiring Knowledge Through Multi-Modal Human-Robot Interaction, 2012–2016

Laurea Magistrale in Electronic Physics

Sapienza, University of Rome

Thesis: Study of the Readout System of a Microcapillary Position Detector Filled with Scintillating Liquid, 2010–2012

Laurea Triennale in Physics

Università degli studi di Pisa

Thesis: Numerical Solution of the Unidimensional Schrödinger Equation, 2007–2010

LANGUAGES

Italian	■ ■ ■ ■ ■
English	■ ■ ■ ■ ■
German	■ ■ ■ ■ ■
French	■ ■ ■ ■ ■

SKILLS

- Agile Development
- Nonviolent Communication
- Effective Leadership
- Python
- C++
- Prolog
- ROS
- Nix, NixOS

ADDITIONAL EXPERIENCE

- **Author, maintainer and contributor of various open and closed source projects**
Repositories: [ros/diagnostics](#), [UP4ROS2](#), [pyswip](#), [PetriNetPlans](#), [ros/geometry2](#), [behavior_trees](#), [robot_state_reasoning](#), [topological_graph_planner](#), and others.
- **European Project Reviewer**
Project: [SciRoc](#) (EU-H2020)
- **SPC, PC and Reviewer for various international conferences**
Conferences: IJCAI, ECAI, AI*IA, and others.
- **International Research Project Contributor**
Projects: AIPlan4EU (EU-H2020 funded project), RoboDevOps (Bavarian funded project)
- **Robocup 2013: Team Leader and coordination and behaviors developer of the SPQR team**
Achieved 1st place at Iran Open 2013 & 3rd place at German Open 2013

Senior Robotics Software Engineer Mar 2017 – Dec 2017 Magazino GmbH | Munich

- Led the development of behaviors, navigation, and deployment for [TORU](#), a state-of-the-art autonomous robot.
- Contributed to TORU's recognition as Best ROS-Based Product in 2020 and Best Product at LogiMAT in 2018.
- Played a key role in advancing TORU's capabilities, driving its adoption in commercial and industrial applications.

Keywords: ROS, Behavior Trees, Django, Diagnostics, Prolog, Error modeling and Handling.

Robotics Software Engineer Nov 2015 – Mar 2017 Magazino GmbH | Munich

- Developed advanced navigation algorithms and behaviors for autonomous robots.
- Optimized robotic systems for improved performance, reliability, and adaptability in dynamic environments.

Keywords: ROS, Move Base, Topological Graph Planner, NetworkX, Django, Behavior Trees.

SELECTED PUBLICATIONS

Patents

- *Testbench for robot behavior trees.* **G. Gemignani**, M. Maerz, C. Ngan. European Patent and Trademark Office, filed on May 7th, 2024.
- *Tasking robots using a job data structure.* **G. Gemignani**, R. Mansilla Martin, M. Grimm, M. Maerz, M. Tenorth, C. Ngan. European Patent and Trademark Office, filed on May 7th, 2024.
- *Capabilities for error categorization, reporting and introspection of a technical apparatus.* **G. Gemignani**, M. Grimm, M. Tenorth. European Patent and Trademark Office, filed on August 8th, 2021.
- *Controlling an apparatus, e.g., a robot, with a behavior tree.* **G. Gemignani**. European Patent and Trademark Office, filed on February 4th, 2021.

Papers

- *Planning for Automated Testing of Implicit Constraints in Behavior Trees.* U. Köckemann, D. Calisi, **G. Gemignani**, J. Renoux, and A. Saffiotti. International Conference on Automated Planning and Scheduling, ICAPS 2023.
- *Dialogue with Robots to Support Symbiotic Autonomy.* A. Vanzo, D. Croce, E. Bastianelli, **G. Gemignani**, R. Basili, D. Nardi. Dialogues with Social Robots, pp. 331-342, 2017.
- *Living with Robots: Interactive Environmental Knowledge Acquisition.* **G. Gemignani**, R. Capobianco, E. Bastianelli, D. D. Bloisi, L. Iocchi, D. Nardi. Robotics and Autonomous Systems, RAS 2016.
- *Multi-Robot Search for a Moving Target: Integrating World Modeling, Task Assignment and Context.* F. Riccio, E. Borzi, **G. Gemignani**, and D. Nardi. International Conference on Intelligent Robots and Systems, IROS 2016. RoboCup Best Paper Award.

[...full list of publications](#)